

UNIT – 5
ASSESSMENT – 1
SOLUTION

1. The system uses two recommendation approaches.

Content-Based Filtering

Recommends items similar to items previously liked by the user.

For example:

User likes Python Course



System identifies similar courses



Recommends Data Science Course

Collaborative Filtering

Finds users with similar preferences and recommends items liked by similar users.

User A and User B have similar preferences



User B likes Item X



Recommend Item X to User A **Hybrid**

Recommendation The final score can be calculated as: where:

α = weight given to content-based recommendation

Intelligent Agent

The agent monitors:

- User clicks
- Ratings
- Purchases
- Likes/dislikes
- Feedback and updates recommendations

accordingly.

2. The system uses PySpark RDDs to process data from multiple sources.

Data Sources

Sensors



Satellite Data



Social Media



PySpark RDD



Data Fusion



Disaster Detection



Intelligent Agent



Alert

Data Fusion

Data from different sources is combined using a common location and disaster type. For example:

Sensor → High water level

Satellite → Flooded area

Social media → "Flood reported"

Together these increase the confidence of flood detection.

Intelligent Agent

The agent:

- Monitors incoming information.
- Calculates disaster confidence.
- Generates warnings.
- Recommends response actions.
- Updates its decision using new observations.

3. A transaction can be considered suspicious when it has abnormal characteristics such as:

Very high amount

+

Unusual location

+

High transaction frequency

↓

Possible Fraud

PySpark RDDs distribute transaction processing across multiple workers.

The intelligent agent monitors suspicious transactions and generates alerts.

4. The system processes inventory and demand data using RDDs.

For example:

Demand data

↓

RDD processing

↓

Inventory analysis

↓

Stock shortage detection

↓

Supply Chain Agent

↓

Restock recommendation

5. The system analyzes student learning data using distributed RDD processing.

Example:

Student Activity



PySpark RDD



Performance Analysis



Weak Topic Detection



Learning Agent



Personalized Recommendation

Intelligent Learning Agent

The agent monitors:

- Scores
- Learning time
- Completed lessons
- Quiz performance
- Feedback

Based on these observations, it recommends appropriate learning material.

